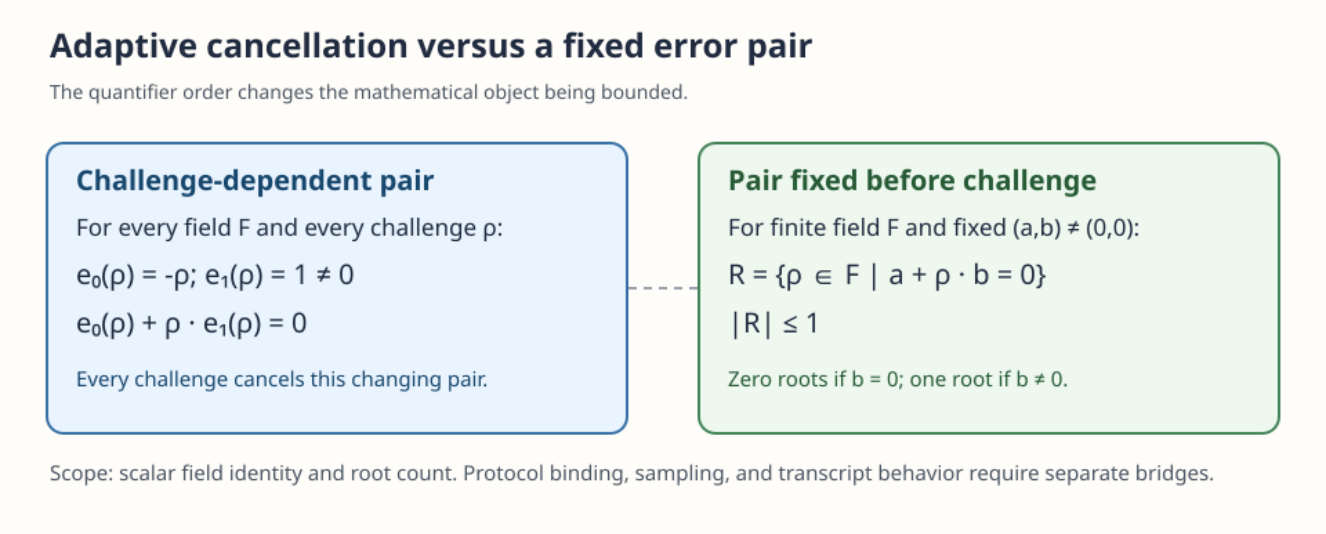
Adaptive batching: scalar cancellation and fixed roots

Formal review packet, 2 November 2026

Claim. Over every field F, the challenge-dependent errors $e_0(\rho) = -\rho$ and $e_1(\rho) = 1$ are nonzero as a pair and satisfy $e_0(\rho) + \rho e_1(\rho) = 0$ for every ρ. For fixed $(a, b) \neq (0, 0)$ over a finite field, the set $\{\rho \in F : a + \rho b = 0\}$ has cardinality at most one.



Evidence. The blind mathematical review checks both claims directly, enumerates $\mathbb{F}_2$, and exhibits the zero-pair and $\mathbb{Z}/4\mathbb{Z}$ boundary cases. The independent source review checks the frozen target and source against the named Lean declaration at commit `8a9daca52daa3c121424bad54c0260cfaedb33b2`. It found and resolved an omitted binder in an earlier head. The exact machine status, receipt identity, and transitive axiom result are recorded below.

Machine status. Hosted run 41488317240 passed on the merge of main `63690685` and proof head `8a9daca5`. Its named receipt for `adaptive_rlc_cancellation` reports `machine_verified`, successful build and axiom audit, and no forbidden constructs. The elaborated type retains both field quantifiers, the finite-field enumeration, and the nonzero fixed-pair premise. Its transitive axioms are `propext`, `Classical.choice`, and `Quot.sound`; the workspace audit counted 1,091 theorems with standard axioms only. The pre/post dependency-pin receipts are byte-identical. The hosted Lean/Lake binaries were not independently rechecked.

Publication gate. The proof PR remains open because two unrelated DREG supersession harness checks failed on the tested merge tree. The repair is separately proposed in PR 2973; the proof must pass its own current-head checks after that repair before merging.

Review process. Both reviewers recorded their reads and scoped verdicts. The repository independence checker rejected this round's prewritten Markdown plan because it requires a YAML `review_plan` block. This procedural check remains open; its failure is not a Lean or mathematical counterexample.

Scope. The theorem concerns a scalar field identity and a univariate linear root count. A probability statement additionally needs a specified challenge distribution and independence from the fixed pair. Applying the identity to a batching protocol needs a verified mapping from protocol errors and acceptance to this equation, including transcript and binding behavior. No protocol-security or performance claim is derived here.

Review artifacts. The source and blind reports, frozen review plan, current-head machine receipt, and source hashes accompany this PDF in the archival packet. The diagram is rendered from the adjacent SVG.